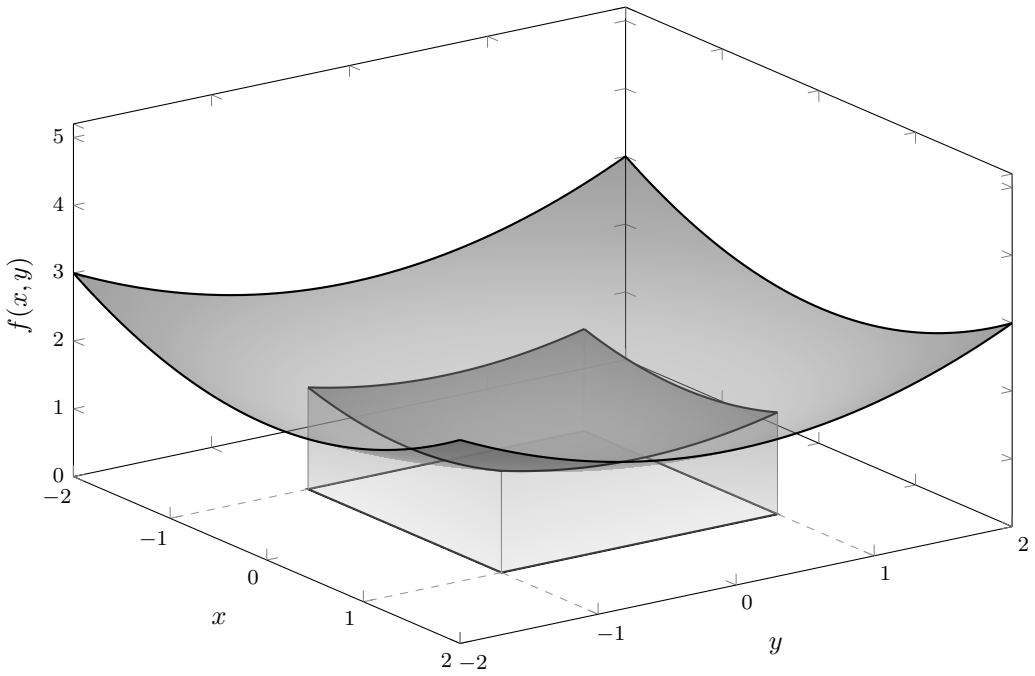
Multiple Integrals

Multiple integrals are a way of generalising integral under the curve to higher dimensions. A double integral is an integral of a function with two variables $f(x, y)$ and is now a volume under a surface rather than an area under a line Consider the diagram below for the function $f(x, y) = \frac{1}{4}(x^2 + y^2) + 1$



The diagram shows the area under the function $f(x, y)$ on the domain $[-1, 1] \times [-1, 1]$. The expression for this integral can be written in the following ways

$$V = \iint_{[-1,1]\times[-1,1]} \left(\frac{1}{4}(x^2 + y^2) + 1\right) dx\, dy$$
$$V = \int_{-1}^{1}\int_{-1}^{1} \left(\frac{1}{4}(x^2 + y^2) + 1\right) dx\, dy$$

Notice there are now two integral symbols, which are there to signify that we integrate with respect to two variables. Alongside the second integral, a dy must be included in addition to the dx. Evaluating double integrals has no major conceptual leaps if you understand integration with one variable and the concept of fixing one variable − the same as from partial derivatives. In order to calculate the definite integral, first calculate the inside integral while treating y as constant. It should be noted that the bounds are to be substituted into the x variable, since the dx is on the inside integral. Secondly, evaluate the outside

integral as you would a regular integral. A demonstration of this process follows

$$\begin{aligned} V &= \int_{-1}^{1}\int_{-1}^{1}\left(\frac{1}{4}(x^2+y^2)+1\right)dx\,dy \\ &= \frac{1}{4}\int_{-1}^{1}\int_{-1}^{1}(x^2+y^2+4)\,dx\,dy \\ &= \frac{1}{4}\int_{-1}^{1}\left[\frac{1}{3}x^3+xy^2+4x\right]_{x=-1}^{x=1}dy \end{aligned}$$

Taking a quick aside to evaluate the integrand given the bounds of x

$$\begin{aligned} \left[\frac{1}{3}x^3+xy^2+4x\right]_{x=-1}^{x=1} &= \left(\frac{1}{3}(1)^3+1\cdot y^2+4\cdot 1\right) \\ &\quad -\left(\frac{1}{3}(-1)^3+(-1)\cdot y^2+4\cdot(-1)\right) \\ &= 2y^2+\frac{26}{3} \end{aligned}$$

Returning to the double integral

$$\begin{aligned} V &= \frac{1}{4}\int_{-1}^{1}\left(2y^2+\frac{26}{3}\right)dy \\ &= \frac{1}{4}\left[\frac{2}{3}y^3+\frac{26}{3}y\right]_{y=-1}^{y=1} \\ &= \frac{1}{4}\left(\frac{2}{3}\cdot 1^3+\frac{26}{3}\cdot 1\right)-\frac{1}{4}\left(\frac{2}{3}\cdot(-1)^3+\frac{26}{3}\cdot(-1)\right) \\ &= \frac{14}{3} \end{aligned}$$

Integrating Over Non-Rectangular Regions

Before integrating non-rectangular domains, we will quickly discuss the concept of the bounds of an integral.

Consider the following integral

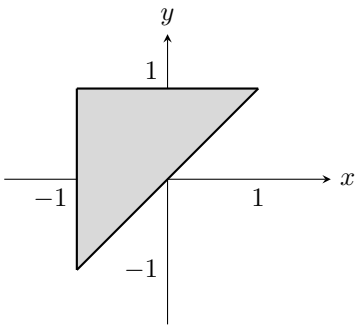
$$\int_{-1}^{1} x^2 \, dx$$

This expression when spoken verbally will typically be read *the integral of* x^2 *from -1 to 1.* It is important to make the conceptual clarification that the integral is being performed *on the region [-1,1].* Likewise, the following double integral

$$V = \int_{x=-2}^{x=0} \int_{y=-1}^{y=1} \frac{1}{4}(x^2 - y^2) \, dy \, dx$$

is being performed over the region $\{-2 \leq x \leq 0, \ -1 \leq y \leq 1\}$. This concept is particularly important when dealing with double integrals because it makes the concept of integrating over non-rectangular domains easier.

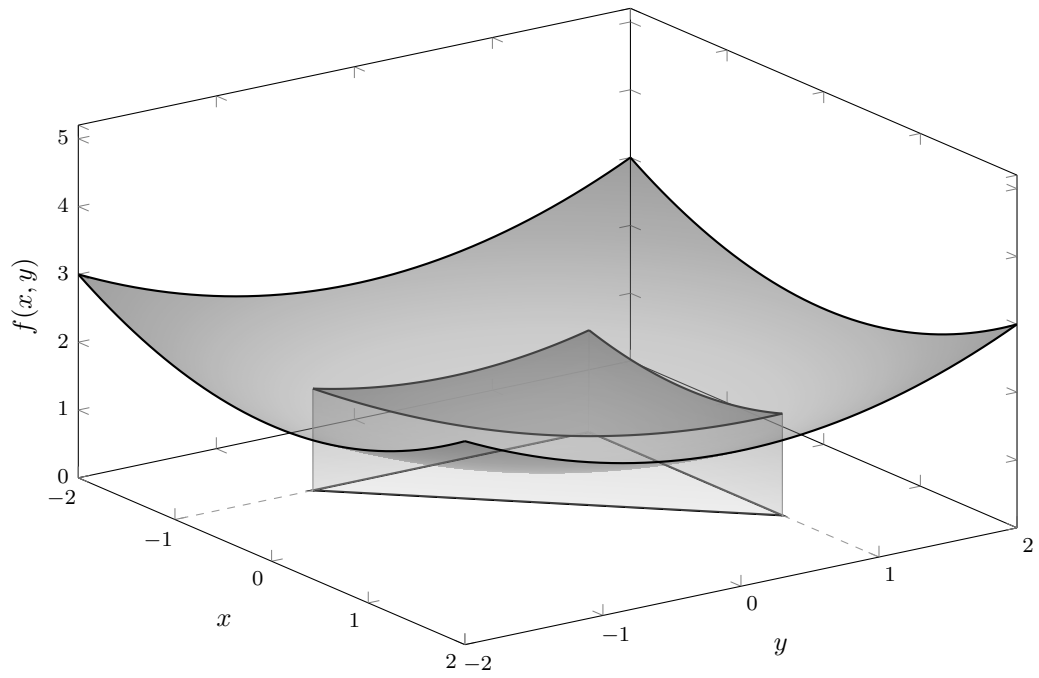
Consider the following region in the $x - y$ plane



Unlike the rectangular region $\{-2 \leq x \leq 0, \ -1 \leq y \leq 1\}$, the upper and lower $y-$bounds cannot both be fixed numbers. The lower $y-$bound changes with x, in particular the highlighted region sits above the line $y = x$, so the expression for the region must rely on using the x variable for lower bound for the y variable, giving $x \leq y$. Given these constraints, the expression for the region is $\{-1 \leq x \leq 1, \ x \leq y \leq 1\}$. So, if we copy the upper and lower bounds given in the expression and use them in the bounds for a double integral below

$$\int_{x=-1}^{x=1} \int_{y=x}^{y=1} \Big(\frac{1}{4}(x^2 + y^2) + 1 \Big) dy \, dx$$

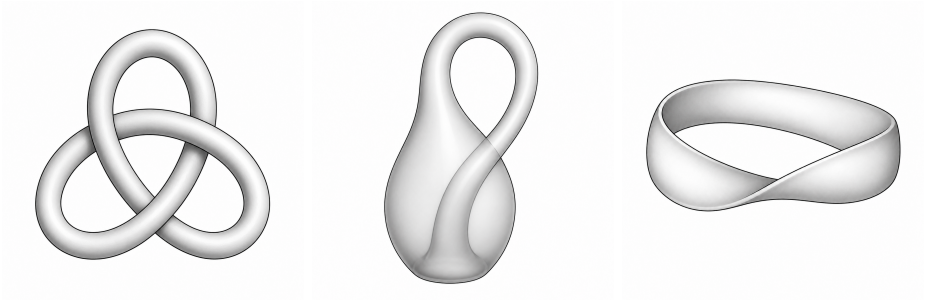
we can integrate our function $f(x, y)$ over a region of this shape



and it evaluates the exact same as the example double integral at the beginning. A keen individual could try calculate this for them self and verify that this equals $\frac{14}{6}$ (half the value of the worked example)

Topology

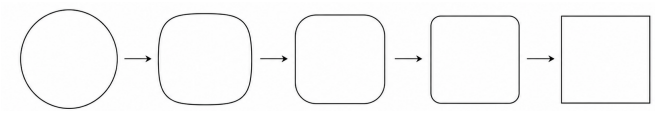
Students who have engaged with but not studied topology typically envision weird shapes, geometric objects and knots.



While these things all show up eventually, the definition of a topological space often stumps students when they first encounter it.

Topology looks at geometry but does so without measurement. The main thing the 'topology' of an object allows us to study is how it behaves when you stretch it. We call two objects *homeomorphic* when you can continuously deform one object into the other - i.e. without creating or filling holes, creating or losing dimensions, gluing itself or passing through itself. It turns out, all the concepts we care about in a strictly topological sense do not require actual measurement. When it comes to looking at holes, knots and self-intersection, the distance between two points doesn't matter. This is because it doesn't matter how big the hole is, we just care that it's there. The only interesting information we care about in topology is the underlying structure of the object, we don't care about about size or distance.

Consider a square and a circle - specifically the boundaries. A circle is defined as the collection of points equidistant from some chosen point, and a square defined as a quadrilateral with four equal length sides. Topologically, these are both the same, because they both have one hole, have an inside and an outside, and do not self intersect. So these two objects are *homeomorphic*, and when we picture a homeomorphism, we picture it like an animation of one object turning into another, as below



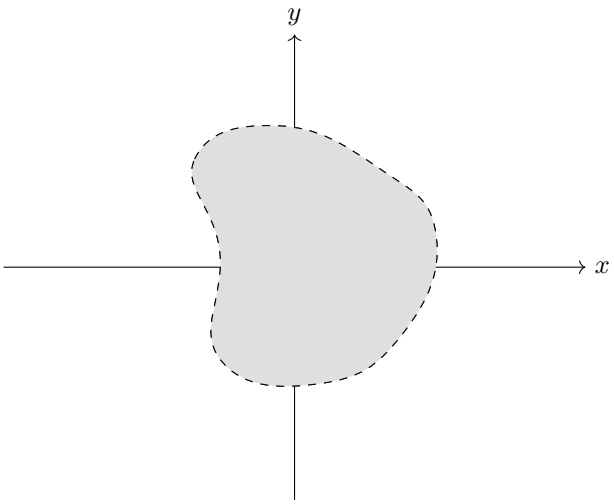
None of these mentioned topological concepts require the concept of a numerical distance.

A lot of the spaces so far in this book - vector spaces, metric spaces, groups, rings - are sets of objects with some sort of structure, whether that be addition, multiplication, scaling, or distance. A topological space has none of this - because it doesn't need it! A topological space is about as bare bones as it gets, basically just a set of points, with a small requirement on the structure so that we can examine closeness (but not numerically).

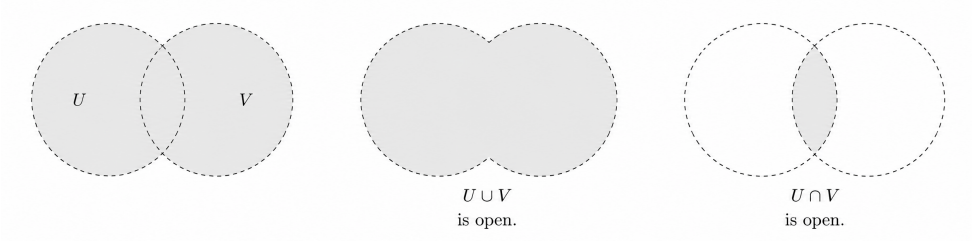
So what is a topological space? In order to understand, we need to look at one concept. Topologically, we still care about the boundaries of our objects, so we want to ensure our extra structure accounts for this. When we look at two sets U and V, there's two common operations we can perform

- Union ($U \cup V$): Returns all the values from U and V
- Intersection ($U \cap V$): Returns the values common to both U *and* V

An open set in $\mathbb{R}$ is an interval where the parentheses are round, e.g $(0, 1)$. Another way to think of open sets is that open sets do not have their boundaries as part of the set. If we were to draw a generic set in the $x-y$ plane, i.e. an open set of $\mathbb{R}^2$, we would represent an open set with a dashed line and a closed set with a solid line. So a diagram of a generic open set on the $x-y$ plane would look like



If we take a union of two open sets or an intersection of two open sets, they will both return open sets.

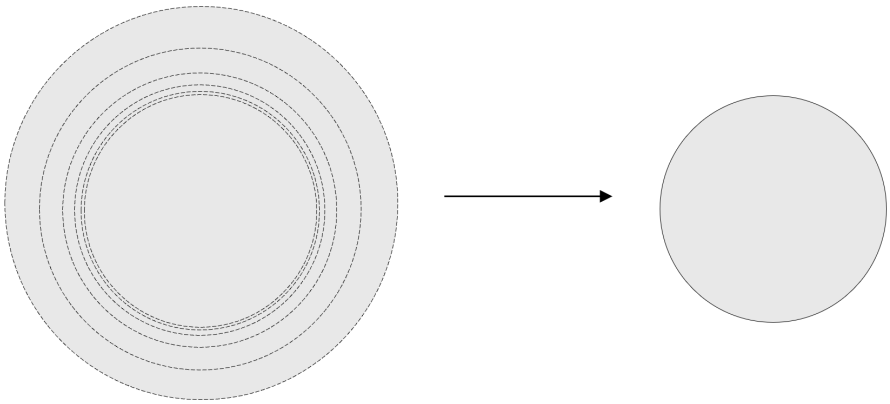


And this is true for a union and intersection to any number of finite open sets. It turns out even a union of an infinite series of sets is open. However, an intersection of an infinite set of open sets *may* be closed.

Consider a series of concentric open circles.

The radius of the first ball is $r_1 = 1 + \frac{1}{1} = 2$, and we'll call this set U_1.
The second circle will have a radius $r_2 = 1 + \frac{1}{2} = \frac{3}{2}$ and we'll call this set U_2.
Similarly, the third circle will have a radius $r_3 = 1 + \frac{1}{3} = 1\frac{1}{3}$ and be called U_3.

This infinite series of concentric circles converges in on a circle of radius $r = 1$.



So the intersection of all these sets consists of points that are inside all the sets U_n. This consists of all the points inside the circle of $r = 1$ including the boundary. In particular, this means the intersection is *not open*, it is *closed*.

Definition of a topological Space

So now that we know this, we can make sense of the definition of a topological space. A topological space is a set of points X, coupled with a *topology*, notated τ, (the greek letter *tau*), which is a list of sets that we declare to be *'open'*. In this list of open sets τ we must have:

- The empty set (notated $\emptyset$) and the whole set X

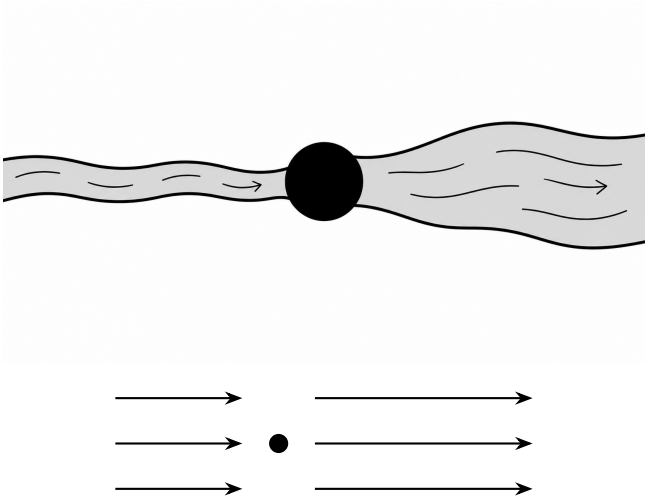
- For any finite or infinite combination of open sets, the union of these sets must also be in the list
- For any finite combination of open sets, the intersection of these sets must also be in the list.

Just like we enforce certain conditions on our distance function to align with our intuition in $\mathbb{R}^n$, we also enforce this condition on the list of open sets to align with our rules of union and intersection of open sets in $\mathbb{R}^n$.

So you can probably see why students get stumped when they have this definition given to them. The definition does not at all look conducive to studying topology as we know it to be. But it turns out this is all the structure we need to study holes, knots, boundaries, self intersection and deformations. But you may ask, why do we need to declare what sets are 'open'? This is to make continuity work. Continuity was discussed earlier, however there is an equivalent definition.

Divergence

There are two notions of the 'derivative' of a vector field, the first being the divergence. Divergence is calculated at each point in a vector field. The divergence at a point can be thought of as a small water hole around that point, where water can flow in and out.



In this water hole analogy, the direction of the vector field represents the direction of flowing water, and the magnitude represents the intensity. The graphic shows a thin stream of water entering the point and a wider stream of water exiting. This is representative of a positive divergence. Using the corresponding vector analogy, a vector field points toward a point and increases in magnitude on the other side of the point, creating a net outwards flow. The vector field at this point has a positive divergence

The divergence at a particular point is the net water flow. The divergence is positive if the net flow results in the hole acting as a water source, where the net flow is out of the hole. The divergence becomes increasingly positive the higher the intensity of the net outwards water flow. Vice versa, a negative divergence means the net water flow acts as a water sink, where the net water flow is into the hole. The magnitude becomes increasingly negative the higher the intensity of the net inwards water flow. The divergence will be zero if the the total outwards water flow exactly offsets the inwards flow, resulting in a net zero flow.

This concept has been presented for a two dimensional vector field, but the concept extends analogously to three dimensions. For a three-dimensional vector field $\mathbf{F}(x, y, z) = (f_x, f_y, f_z)$, the formula for divergence is

$$\mathbf{div}(\mathbf{F}(x, y, z)) = \frac{\partial f_x}{\partial x} + \frac{\partial f_y}{\partial y} + \frac{\partial f_z}{\partial z}$$

The alternate form involves the use of the grad operator ∇. We have seen ∇ used for $\nabla f(\mathbf{x})$ to calculate the direction of steepest ascent at the point $\mathbf{x}$. We have also seen ∇ used for ∇f, to give the vector field where each vector at each point corresponds to the direction of steepest ascent. In addition to this, mathematicians will also use the symbol ∇ on its own as the vector

$$\begin{pmatrix} \dfrac{\partial}{\partial x} \\ \dfrac{\partial}{\partial y} \\ \dfrac{\partial}{\partial z} \end{pmatrix}$$

which is neither a vector or a vector field. However, ∇ on its own can be used in conjunction with the vector field

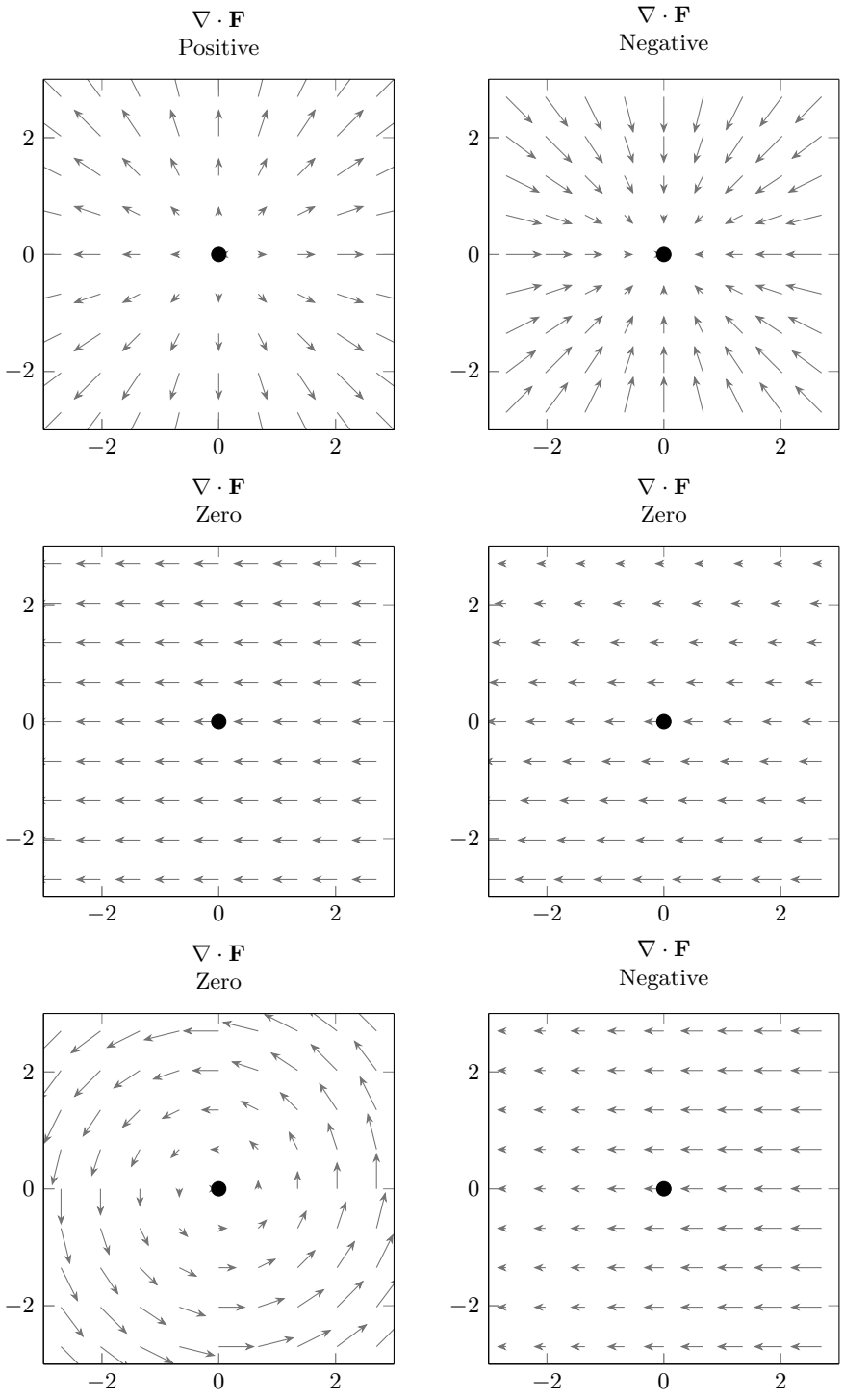
$$\mathbf{F} = \begin{pmatrix} f_x \\ f_y \\ f_z \end{pmatrix}$$

and together with a dot product they produce

$$\nabla \cdot \mathbf{F} = \begin{pmatrix} \dfrac{\partial}{\partial x} \\ \dfrac{\partial}{\partial y} \\ \dfrac{\partial}{\partial z} \end{pmatrix} \cdot \begin{pmatrix} f_x \\ f_y \\ f_z \end{pmatrix} = \frac{\partial f_x}{\partial x} + \frac{\partial f_y}{\partial y} + \frac{\partial f_z}{\partial z}$$

This convention is often why the formula for divergence is written as

$$\mathbf{div}(\mathbf{F}) = \nabla \cdot \mathbf{F}$$


$\nabla \cdot \mathbf{F}$
Positive
$\nabla \cdot \mathbf{F}$
Negative
$\nabla \cdot \mathbf{F}$
Zero
$\nabla \cdot \mathbf{F}$
Zero
$\nabla \cdot \mathbf{F}$
Zero
$\nabla \cdot \mathbf{F}$
Negative
2
0
−2
−2
0
2

Calculating the divergence at each point will output a regular scalar function $f : \mathbb{R}^2 \to \mathbb{R}$, which you can visualise using a heat map, or in the case of two dimensional vector fields, a surface in 3D.

